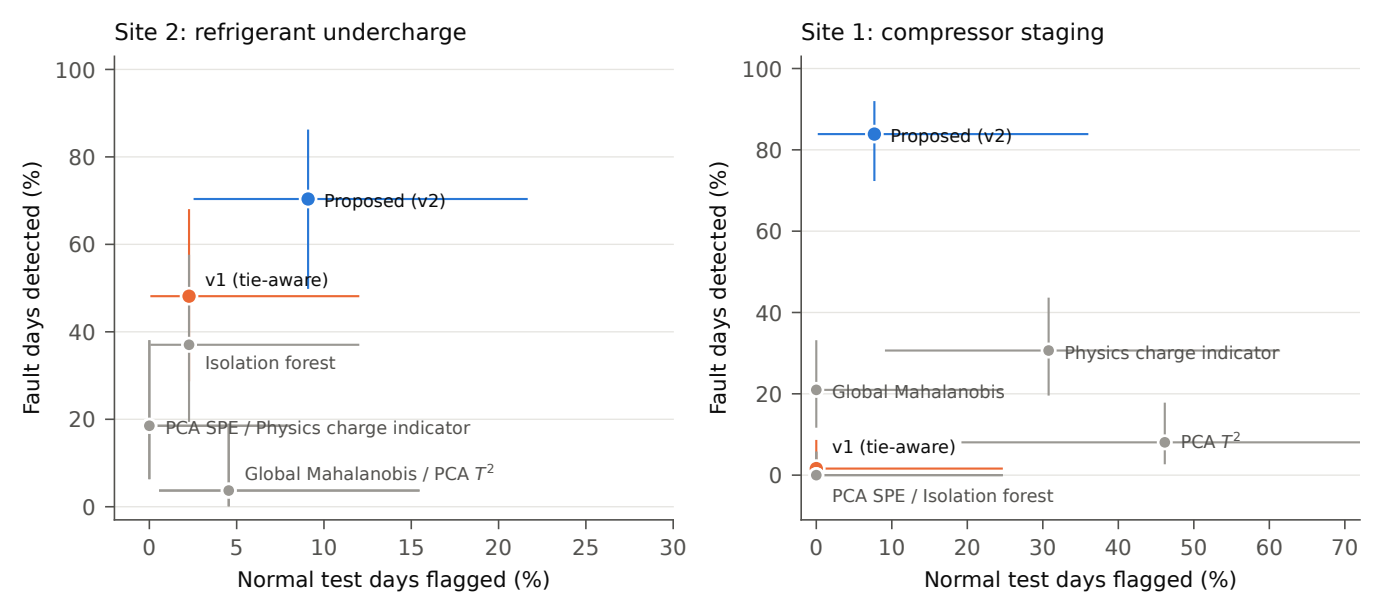
Site 2: refrigerant undercharge
Fault days detected (%)
Normal test days flagged (%)
Proposed (v2)
v1 (tie-aware)
Isolation forest
PCA SPE / Physics charge indicator
Global Mahalanobis / PCA T^2
Site 1: compressor staging
Fault days detected (%)
Normal test days flagged (%)
Proposed (v2)
Physics charge indicator
Global Mahalanobis
v1 (tie-aware)
PCA T^2
PCA SPE / Isolation forest